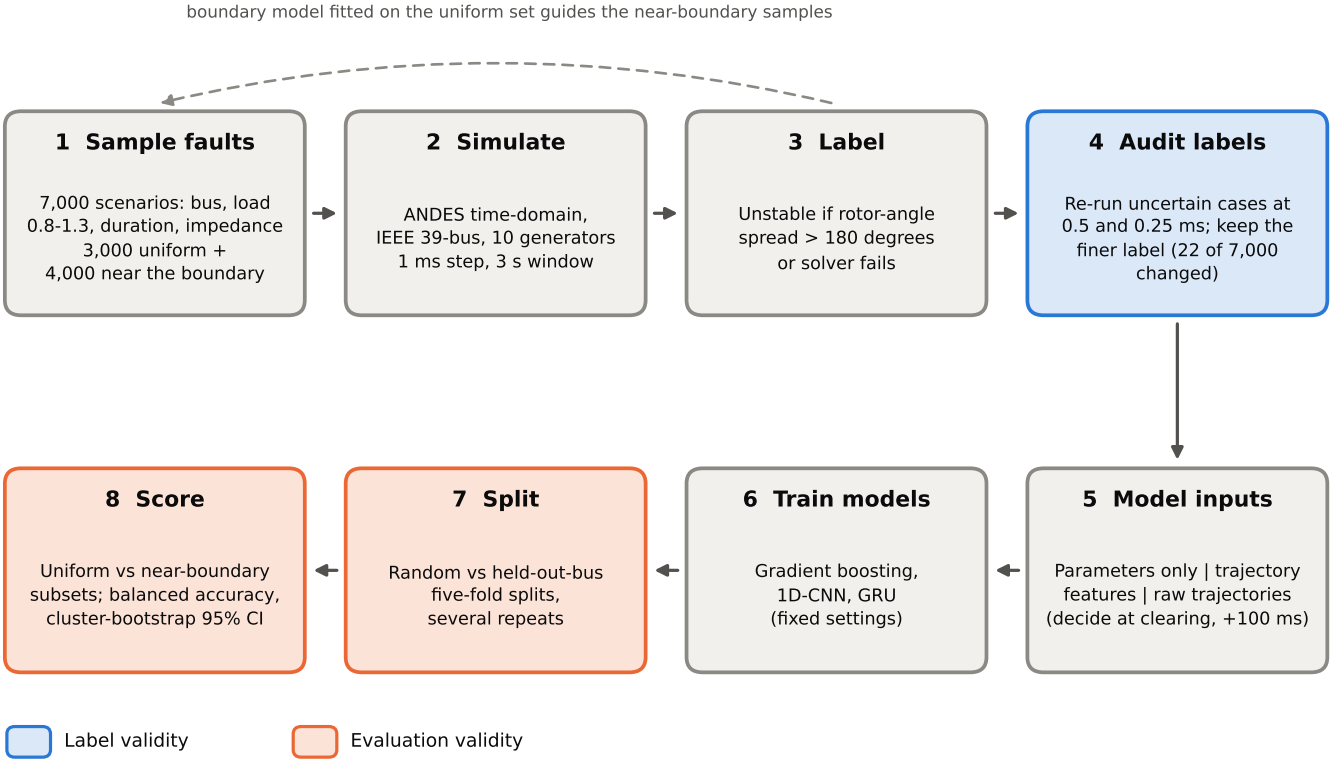
boundary model fitted on the uniform set guides the near-boundary samples
1 Sample faults
7,000 scenarios: bus, load 0.8-1.3, duration, impedance
3,000 uniform + 4,000 near the boundary
2 Simulate
ANDES time-domain, IEEE 39-bus, 10 generators
1 ms step, 3 s window
3 Label
Unstable if rotor-angle spread > 180 degrees or solver fails
4 Audit labels
Re-run uncertain cases at 0.5 and 0.25 ms; keep the finer label (22 of 7,000 changed)
5 Model inputs
Parameters only | trajectory features | raw trajectories
(decide at clearing, +100 ms)
6 Train models
Gradient boosting, 1D-CNN, GRU
(fixed settings)
7 Split
Random vs held-out-bus five-fold splits, several repeats
8 Score
Uniform vs near-boundary subsets; balanced accuracy, cluster-bootstrap 95% CI
Label validity
Evaluation validity